Homework 1

Due : Sunday, April 12 at 11 : 59 pm

Honor Code

Write and sign the following statement:

> "I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

1 Online K-Means

Consider the sequential (online) version of the K-means algorithm. Online algorithms assume that you receive data points in a *stream* rather than having all of your data already stored.

In online K-means, instead of using the full dataset to update cluster centers at each iteration, we update the nearest prototype μ_k for each data point x_n as it is observed, using the rule:

$$\mu_k \leftarrow \mu_k + \frac{1}{N_k}(x_n - \mu_k),$$

where N_k is the number of points that have been assigned to cluster k so far. Each point is assigned to the cluster with the nearest center at the time it is observed.

(a) Explain how this sequential update differs from batch K-means in terms of memory and computation. *Hint: How often is each data point used in regular K-means compared to online K-means?*

(b) Show that after each update, the objective function

$$E(\{\mu_k\}) = \sum_{n=1}^{N} \min_k \|x_n - \mu_k\|^2$$

does not increase.

2 MLE

Suppose we have a coin with unknown bias Y. If we flip the coin we observe a Bernoulli random variable:

$$X \sim \textbf{Bernoulli}(Y) \;=\; p(X = x \mid Y) = Y^x(1-Y)^{1-x}$$

The coin will land heads ($X = 1$) with probability Y and tails ($X = 0$) with probability $1 - Y$. We flip the coin (independently) N times to collect the dataset $\mathcal{D} = \{X_n\}_{n=1}^N$. We observe the coin lands heads $N_1 = \sum_{n=1}^N X_n$ times and tails $N_0 = N - N_1$ times.

For this problem, we will also introduce a prior on the probability Y that the coin lands heads. The Beta distribution is a common prior for probabilities. The Beta distribution has the density:

$$p(y \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1}(1-y)^{\beta-1}, \quad y \in (0, 1),$$

where $\alpha > 0$ and $\beta > 0$ are parameters of the distribution. The function $B(\alpha, \beta)$ is the normalization function

$$B(\alpha, \beta) = \int_0^1 y^{\alpha-1}(1-y)^{\beta-1} dy$$

of the Beta distribution called the Beta function. As a prior we will need to pick values for α and β. These should be based on our beliefs about the coin. Here we will use $\alpha = 3$ and $\beta = 3$:

(a) Write the likelihood $p(\mathcal{D}|Y)$ of the data given Y.

(b) Compute the derivative of the log-likelihood function with respect to Y and solve for the MLE estimate for Y.

(c) Calculate the bias of the MLE estimate.

3 Moments of Mixture Models

Consider a mixture distribution of the form

$$p(x) = \sum_{k=1}^K \pi_k p(x|k)$$

where the elements of x could be discrete or continuous or a combination of these. Denote the mean and covariance of $p(x|k)$ by μ_k and $\sum_k$ respectively.

(a) Show that

$$E[x] = \sum_{k=1}^K \pi_k \mu_k$$

(b) Show that

$$Cov[x] = \sum_{k=1}^K \pi_k \left(\Sigma_k + \mu_k \mu_k^T\right) - E[x]E[x]^T$$

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1) Online K-Means

a)

Memory

Batch K-Means: Requires the entire dataset to be stored in memory simultaneously. Since every iteration involves calculating distances between all data points and all cluster centers, the memory complexity is $O(N)$. For very large datasets, this can exceed hardware limits.

Online K-Means: Only requires storing the cluster centers and the counts of assigned points. Once a data point x_n is processed to update the nearest center, it can be discarded. The memory complexity is reduced to $O(k)$, making it ideal for streaming data.

Computation

Batch K-Means: Every data point is used multiple times until the algoritm converges. This leads to high overhead for large N.

Online K-Means: Each data point is used exactly once to perform a single update. There are no global iterations over the full dataset. This significantly reduces the computational burden per time step and allows the model to adapt in real-time as new data arrives.

b)

To show that $E(\{\mu_k\})$ does not increase after an update, we can look at the update rule from two perspectives:

Gradient Descent Perspective

The objective function is defined as:

$$E = \sum_{n=1}^{N} \|x_n - \mu_{k(n)}\|^2$$

where $k(n)$ is the index of the cluster nearest to x_n. To minimize this with respect to a specific center μ_k, we take the negative gradient:

$$-\frac{\partial E}{\partial \mu_k} = 2 \sum_{n \in \text{cluster } k} (x_n - \mu_k)$$

The online update rule $\mu_k \leftarrow \mu_k + \frac{1}{N_k}(x_n - \mu_k)$ is a stochastic gradient descent step with a learning rate $\eta = \frac{1}{N_k}$. Since we are moving μ_k in the direction of the negative gradient (towards the new data point x_n), the value of the quadratic error function E for that specific point decreases or stays the same.

Minimization via the Mean

Mathematically, the value μ that minimizes the sum of squared distance $\sum \|x_i - \mu\|^2$ for a set of points is the arithmetic mean of those points.

Let μ_k^{old} be the mean of the first $N_k - 1$ points

When the N_k-th point (x_n) arrives, the update rule:

$$\mu_k^{new} = \mu_k^{old} + \frac{1}{N_k}(x_n - \mu_k^{old}) = \frac{(N_k - 1)\mu_k^{old} + x_n}{N_k}$$

This formula calculates the new arithmetic mean of all N_k points. By definition, the new mean μ_k^{new} is the optimal point that minimizes the squared error for the current set of points assigned to cluste k.

There foce:

$$\sum_{i=1}^{N_k} \|x_i - \mu_k^{new}\|^2 \leq \sum_{i=1}^{N_k} \|x_i - \mu_k^{old}\|^2$$

Since only one center μ_k changes and it moves to a position that is more optimal for its members, the total objective function E does not increase.

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2 MLE

a) The dataset D consist of N independent and identically distributed Bernoulli trials. The likehood of the given the parameter Y is the product of individual probabilities:

$$p(D|Y) = \prod_{n=1}^{N} p(X_n|Y) = \prod_{n=1}^{N} Y^{X_n}(1-Y)^{1-X_n}$$

by combining the exponents:

$$p(D|Y) = Y^{\sum X_n}(1-Y)^{\sum(1-X_n)}$$

Substituting $\sum X_n = N_1$ (total heads) and $N - N_1 = N_0$ (total tails), we get:

$$p(D|Y) = Y^{N_1}(1-Y)^{N_0}$$

b) To find the MLE, it is easier to maximize the natural logarithm of the likelihood function, denoted as $\ell(Y)$:

$$\ell(Y) = \ln(Y^{N_1}(1-Y)^{N_0}) = N_1 \ln Y + N_0 \ln(1-Y)$$

$$\frac{\partial \ell}{\partial \gamma} = \frac{N_1}{\gamma} - \frac{N_0}{1-\gamma}$$

Setting the derivative to zero to find the maximum:

$$\frac{N_1}{\hat{\gamma}} - \frac{N_0}{1-\hat{\gamma}} = 0 \Rightarrow \frac{N_1}{\hat{\gamma}} = \frac{N_0}{\hat{\gamma}}$$

$$N_1(1-\hat{\gamma}) = N_0\hat{\gamma} \Rightarrow N_1 - N_1\hat{\gamma} = N_0\hat{\gamma}$$

$$N_1 = \hat{\gamma}(N_1 + N_0)$$

since $N = N_1 + N_0$, the MLE estimate is:

$$\hat{\gamma}_{MLE} = \frac{N_1}{N}$$

c) The bias of an estimator is defined as

$$Bias(\hat{\gamma}) = E[\hat{\gamma}] - \gamma.$$

Since N_1 follows a Binomial distribution $B(N, \gamma)$, its expected value is $E[N_1] = N\gamma$.

$$E[\hat{\gamma}] = E\left[\frac{N_1}{N}\right] = \frac{1}{N}E[N_1] = \frac{1}{N}(N\gamma) = \gamma$$

$$Bias(\hat{\gamma}) = \gamma - \gamma = 0$$

The MLE estimate for the Bernoulli distribution is unbiased.

3) Moments of Mixture Models

a) The expected value $E[x]$ is defined by the integral of x weighted by its density $p(x)$:

$$E[x] = \int x p(x) dx$$

Substitute the mixture distribution $p(x) = \sum_{k=1}^{K} \pi_k p(x|k)$ into the equation:

$$E[x] = \int x \left(\sum_{k=1}^{K} \pi_k p(x|k) \right) dx$$

Since the integral is linear, we can swap the summation and the integral:

$$E[x] = \sum_{k=1}^{K} \pi_k \int x p(x|k) \, dx$$

By definition $\int x p(x|k) \, dx$ is the mean of the k-th component, which is μ_k. Therefore:

$$E[x] = \sum_{k=1}^{K} \pi_k \mu_k$$

b) Using the same substitution logic as in part(a):

$$E[x x^T] = \int x x^T p(x) \, dx = \int x x^T \left(\sum_{k=1}^{K} \pi_k p(x|k) \right) dx$$

$$E[x x^T] = \sum_{k=1}^{K} \pi_k \int x x^T p(x|k) \, dx$$

The term $\int x x^T p(x|k) \, dx$ is the second moment of the k-th component. We know that for any distribution $Cov[x|k] = E[x x^T | k] - \mu_k \mu_k^T$. Thus:

$$E[x x^T | k] = \Sigma_k + \mu_k \mu_k^T$$

Substitute this back into the sum:

$$E[x x^T] = \sum_{k=1}^{K} \pi_k (\Sigma_k + \mu_k \mu_k^T)$$

Plug $E[x x^T]$ into the covariance formula:

$$\mathrm{Cov}(x) = E[xx^T] - E[x]\,E[x]^T$$

$$\mathrm{Cov}(x) = \sum_{k=1}^{K} \pi_k \left(\Sigma_k + \mu_k \mu_k^T\right) - E[x]\,E[x]^T$$